

Type

Attribute

Syntax

```
<Path>.ShiftPlan:table
```

Assignment Value

You can assign a value of data type *table*.

Example

```
MyShiftCalendar.ShiftPlan := shiftTimesTable
```

```
// This code modifies the shift table
var t:table := ShiftCalendar.ShiftPlan
t["From", 1] := 3:00:00
ShiftCalendar.ShiftPlan := t
```

See also

[Tab Shift Times](#)

LockoutZone

LockoutZone

Use the object *LockoutZone* for combining a group of material flow objects. If one of these stations fails, all other stations within the lockout zone stop processing their parts as well.

Image

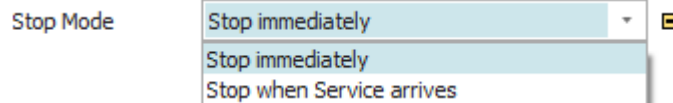


Description

The *LockoutZone* controls the failures of all stations. It then returns the total availability of the assigned stations.

You can insert more than one *LockoutZone* into a simulation model. The *LockoutZones* can overlap, meaning that a station can be assigned to more than one *LockoutZone*.

You have to define *failure profiles* for at least one of the stations assigned to the *LockoutZone*. As soon as one of the stations fails, the *LockoutZone* stops all processing operations of all assigned stations, meaning it sets their attribute **Stopped** to true. You can select if the *LockoutZone* stops the processing operations immediately or when the service arrives.



The stations only start processing *parts* again after all failures were removed. They then only use up the respective remaining processing time.

You can also determine what happens if the stations start processing *parts* again by programming a **Resume Control**.

In case one of the stations is **paused** if the *LockoutZone* intends to stop, this station does not start or restart processing after the pause is over.

You can assign any of the built-in **Material Flow Objects** or **Fluid Objects** to the *LockoutZone*. For stations, you yourself modeled in a *Frame*, the *LockoutZone* sets the attribute **Stopped** of that *Frame* to true. You then have to model an adequate reaction to the attribute. In addition, you can also add a *Worker* or the *WorkerPool*.

If the station that triggers the lockout zone is assigned to more than one *LockoutZone*, all stations of all *LockoutZones* will be stopped. If the station that triggers the lockout zone is assigned to only one *LockoutZone*, it only stops the station that you assigned to it.

To show a **tooltip** with information about the *LockoutZone*, hover with the mouse over it.

To change the length of the graphic and the anchor points of the *LockoutZone*, click **Show Manipulators**



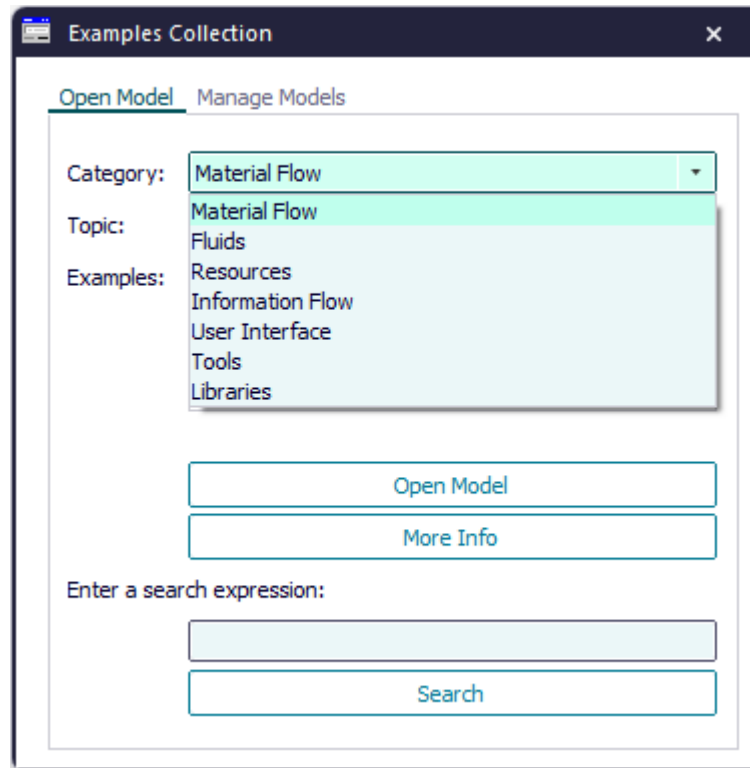
on the **Edit** ribbon tab or press **M** on the keyboard.

Add the Object to the Simulation Model



To add the object *LockoutZone* to your simulation model, click **Manage Class Library > Basic Objects > Resources > LockoutZone** on the **Home** ribbon tab.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



See also

[Model a Lockout Zone](#)

Dialog Box of the LockoutZone

Dialog Box of the LockoutZone

Double-click the icon of the *LockoutZone* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under [Dialog Items of the Objects](#).

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box [Edit 3D Properties](#):

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click **Show Manipulators** on the **Edit** ribbon tab or press **M** on the keyboard.

See also

Active [check box] - LockoutZone

Active [check box] - LockoutZone

To activate the *LockoutZone*, select this check box. It then creates failures and stops the processing operations of all assigned objects, when one of these stations fails. To deactivate it, clear the check box.

☒ Active ☐

SimTalk

Active [SimTalk] - LockoutZone

See also

Objects [LockoutZone]

Tab Controls

Tab Controls [LockoutZone]

Provides controls to modify the built-in behavior of the object.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

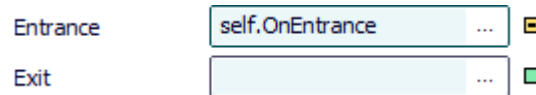
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select [Create Control \[context menu\]](#). *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

Specifying a control overrules the default behavior of the *LockoutZone*. This means that you yourself are responsible for stopping all processing operations of the assigned objects by assigning `true` to their attribute *Stopped*.

When you want the objects to continue processing parts, you have to make sure to terminate **stopped** mode by assigning `false` to the attribute *Stopped*.

See also

[Stopped \[SimTalk\] - material flow objects](#)

[Stop Control \[LockoutZone\]](#)

[Resume Control \[LockoutZone\]](#)

[Stop Mode \[drop-down list\]](#)

Stop Control [LockoutZone]

Modifies the built-in behavior of the object. The object calls the **Stop Control** if one of the stations assigned to the *LockoutZone* fails and thus stops processing parts.

Stop	self.OnStop	...	
Resume	self.OnResume	...	

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.

Entrance		...	☒	☐ Before Actions	☒
Exit		...	☒		☒

Select Object
 Create Control

Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

The anonymous identifier @ designates the triggering station, the anonymous identifier ? designates the *LockoutZone*.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

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Entrance	A1Ctrl	...	☒	☐ Before Actions	☒
Exit		...	☒		☒

Select Object
 Create Control

- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.

Entrance	self.OnEntrance	...	☒
Exit		...	☒

Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

SimTalk

StopCtrl [SimTalk]

See also

Use a Stop Processing Control

Resume Control [LockoutZone]

Modifies the built-in behavior of the object. The object calls the **Resume Control** when all failures of the assigned stations were removed and the stations can thus resume processing *parts*.

Resume	self.OnResume	...	
Stop Mode	Stop immediately	▼	

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.

Entrance		...	☒	☐ Before Actions	☒
Exit		...			☒

Select Object
 Create Control

Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select **Create Control [context menu]**. *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.

Entrance	A1Ctrl	...	☒	☐ Before Actions	☒
Exit		...			☒

Select Object
 Create Control

- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.

Entrance	self.OnEntrance	...	
Exit		...	

Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.

- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

SimTalk

ResumeCtrl [SimTalk]

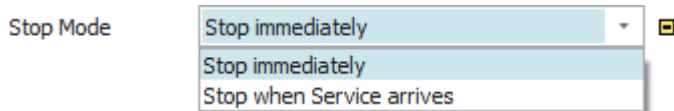
See also

Select Object [for controls]

Use a Resume Processing Control

Stop Mode [drop-down list]

Select the **Stop Mode** of the *LockoutZone*.



Remarks

You can select one of these settings:

- **Stop immediately**

Stops the processing operations of all stations that are part of the *LockoutZone* as soon as one of the stations assigned to it fails. None of the other stations in the model are stopped!

Within this period of time additional failures can take place for the assigned stations, several failures of differing stations can thus overlap. The stations only start processing *parts* again after all failures were removed. They then only use up the remaining processing time.

- **Stop when Service arrives**

Only stops the stations assigned to the *LockoutZone* when the repair services, which the failed station requested, are assigned. Depending on your modeling situation this can be at a different point in time. If you model *Footpaths* on which the *Worker* walks to the station, *Plant Simulation* considers the *Worker* as received once he has reached the station. For *Exporters/Workers* who can be beamed, the service is considered to be received once the *Broker* has assigned the service. This matches the behavior of the **Receive Control**.

The *LockoutZone* does not affect the **Recovery Time** and the **Cycle Time** of the stations which it controls. The *LockoutZone* stops these objects and records statistics values for the state **Stopped**. If you deactivate the *LockoutZone* while it is in the process of stopping other stations, *Plant Simulation* immediately releases all objects stopped by the *LockoutZone*. If you reset your model, all objects change their state from **stopped** to **operational**.

SimTalk

StopMode [SimTalk]

See also

Use a Stop Processing Control

Stopped [state, material flow objects]

Stopped [SimTalk] - LockoutZone

Stop the Associated Stations Immediately After a Failure

Stop the Associated Stations When the Repair Man Arrives

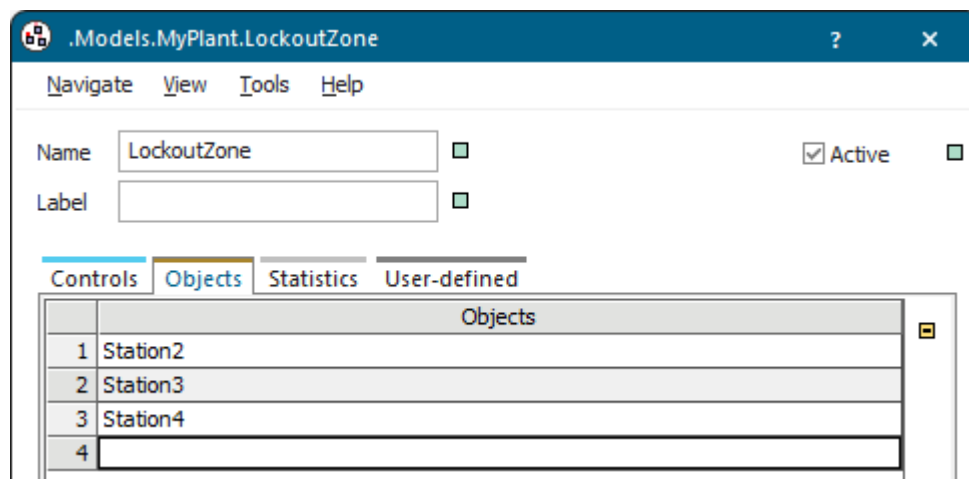
Tab Objects

Tab Objects

Assign resources to the *LockoutZone* on the tab **Objects**.

Remarks

A resource is any of the built-in **Material Flow Objects**, or a *Frame* in which you modeled a machine, whose **Working** hours you would like to control with the *LockoutZone*. In addition, you can also add a *Worker* or the *WorkerPool*.



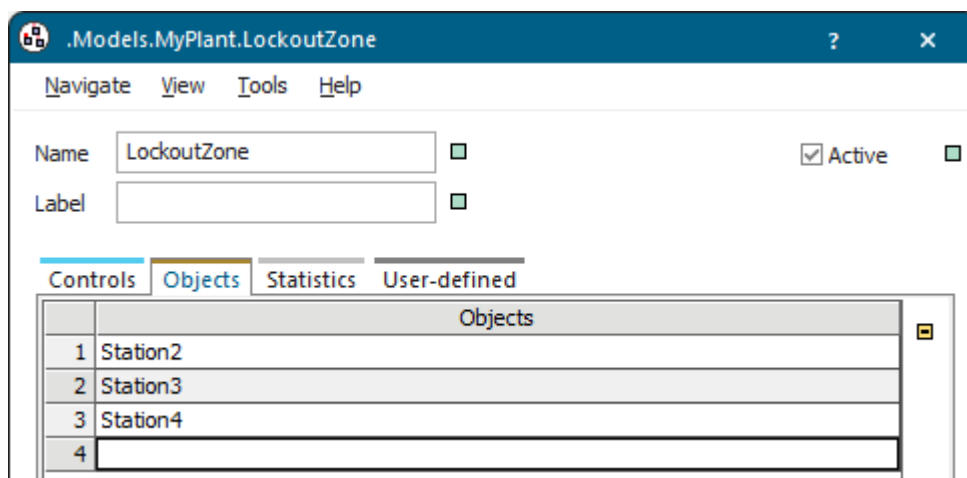
You can manipulate the contents of the list window with the commands of the **Context Menu of Embedded Lists**.

See also

Objects [LockoutZone]

Objects [LockoutZone]

Assign resources, i.e., material flow objects or fluid objects, to the *LockoutZone*.



Remarks

Proceed as follows:

- Before you can type in data, click the **Inheritance** check box so that it looks like this .
- Type the path to and the name of the resource object in a cell. Or
- Drag the resource object to the icon of the *LockoutZone* and drop it there. You can also select and then drag-and-drop multiple objects at the same time!

You can manipulate the contents of the list window with the commands of the **Context Menu of Embedded Lists**.

SimTalk

addObject [SimTalk] - LockoutZone

Objects [SimTalk] - LockoutZone

See also

Show Assigned Objects [LockoutZone]

Tab Statistics [LockoutZone]

The tab **Statistics** shows the most important statistical data of the *LockoutZone*.

Remarks

Item English	Description	Read-only attribute	Item German
Stopped Portion	Shows the portion of the statistics collection period during which the stations were stopped by a <i>LockoutZone</i> in percent.	StatStoppedPortion [SimTalk] - LockoutZone	Angehalten Anteil
Stopped Count	Shows how often the <i>LockoutZone</i> stopped the stations.	StatStoppedCount [SimTalk] - LockoutZone	Angehalten Anzahl
Stopped durations total time	Shows the total time during which the <i>LockoutZone</i> stopped the processing operations of the assigned stations.	StatStoppedTime [SimTalk]	Angehalten dauern Gesamtzeit
Stopped durations mean time	Shows the mean time during which the <i>LockoutZone</i> stopped the processing operations of the assigned stations.	StatStoppedMean [SimTalk] - LockoutZone	Angehalten dauern mittlere Zeit
Stopped durations Standard Deviation	Shows the standard deviation of the time during which the <i>LockoutZone</i> stopped the processing operations of the assigned stations.	StatStoppedDelta [SimTalk] - LockoutZone	Angehalten dauern Standardabweichung
Stopped intervals mean time	Shows the mean time of the intervals during which the <i>LockoutZone</i> stopped the processing operations of the assigned stations.	StatStoppedIntervalMean [SimTalk]	Angehalten Abstand mittlere Zeit
Stopped intervals Standard Deviation	Shows the standard deviation of the intervals during which the <i>LockoutZone</i> stopped the processing operations of the assigned stations.	StatStoppedIntervalDelta [SimTalk]	Angehalten Abstand Standardabweichung

Controls

Objects

Statistics

User-defined

Stopped

Portion15.50%

Total Time11:21:51:28.0886

Count1107

	Mean Time	Standard Deviation
Interval	1:24:24.6735	1:23:24.0204
Duration	15:29.6189	11:00.8606

The **Statistics Report** shows the **Stopped Time** of the individual material flow objects.

Tab User-defined

Define your own attributes as described under the [Tab User-defined](#).

Navigate Menu

The commands are described under the [Navigate Menu](#).

View Menu

View Menu [LockoutZone]

The **View Menu** provides commands to access its functions.

[Refresh \[on View menu\]](#)

[Show Assigned Objects \[LockoutZone\]](#)

[Show Attributes and Methods \[on View menu\]](#)

SimTalk

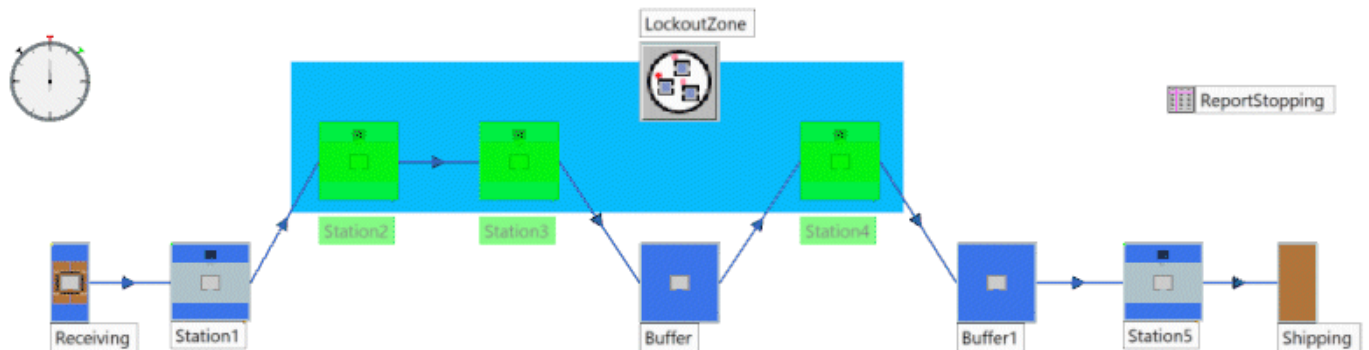
[updateDialog \[SimTalk\]](#)

See also

[View Menu \[general description\]](#)

Show Assigned Objects [LockoutZone]

Selects the objects, which are assigned to the *LockoutZone* as resources in the *Frame* window.



Instead, you can also select **Show Assigned Objects [in Frame]** on the context menu of the *LockoutZone* in the *Frame*.

Tools Menu

The **Tools Menu** provides these menu commands:

Edit Controls

Edit Observers

Help Menu

The commands are described under the [Help Menu](#).

Methods of the LockoutZone

Methods of the LockoutZone

The *LockoutZone* provides:

- The method [addObject \[SimTalk\] - LockoutZone](#).
- The [Methods of All Objects](#).

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window [Show Attributes and Methods](#). The figure below illustrates the information using the example of the object *Station*.

Name	Value	Inherited	Watchable	Signature
contentsList	[]			((Contents:table)) -> any
createAttr				(NameOfUserDefinedAttribute:string, DataType:string) -> boolean
createIcon				(IconName:string, Width:integer, Height:integer) -> integer
deleteAttr				(NameOfUserDefinedAttribute:string) -> boolean

Name of the method Identifier and data type of the parameter you enter Data type of the return value

Name	Value	Inherited	Watchable	Signature
moveToFolder				(Folder:object) -> object
MU	VOID			([Index:integer:=1]) -> object
MUPart	VOID			([Index:integer:=1]) -> object

Name of the method Identifier and data type of the parameter you enter Default value Data type of the return value

- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected [Instance \[general description\]](#).